

HARSHIT RAJ RATAN SINGH

Software Engineer Intern

☎ +91-639630117

@ hrjratansingh@gmail.com

📍 Varanasi - India

in [Linkedin](#)

🐙 [Github](#)

[Leetcode](#)

EXPERIENCE

Software Development Intern

AmasQis.ai

📅 July 2025 – Dec 2025

📍 Remote

- Built high-performance **Node.js/Express.js** microservices, optimizing PostgreSQL and MongoDB queries to reduce API latency by **44%**
- Configured GitHub Actions CI/CD pipelines, reducing release time by **80%** and deployment errors by **80%**.
- Integrated LLMs, RAG, and vector databases, reducing retrieval latency by **50%** and improving response accuracy by **38%**.
- **Containerized** backend services with **Docker** and **Docker Compose**, reducing setup time by **80%** and image size by **35%**.

Frontend Development Intern

ITJOBX

📅 Mar 2026 – May 2026

📍 Remote

- Developed and maintained **5+ responsive** React.js modules using HTML, CSS, JavaScript, and MySQL, reducing page load time by **35%** (2.0s → 1.3s)
- Integrated 10+ RESTful APIs for authentication and CRUD operations, reducing data fetch latency by **40%** (250 ms → 150 ms).
- Optimized React performance using useMemo and useCallback, cutting unnecessary re-renders by **45%** and improving UI response time from **180 ms to 95 ms**.

TECHNICAL SKILLS

- **Languages:** Java, Python, C++, JavaScript, TypeScript
- **FullStack:** React.js, Next.js, Node.js, Express.js, HTML5, CSS3, Tailwind CSS, REST APIs, MongoDB, MySQL, PostgreSQL, Framer Motion, Three.js
- **AI / ML:** GenAI, LLMs, ML, NumPy, Pandas, Matplotlib, Seaborn, Probability and Statistics, Linear Algebra, Langchain, Langgraph, Vectordb
- **Tools:** Git, GitHub, Postman, VS Code, IntelliJ IDEA, Jupyter Notebook, Basic CI/CD (GitHub Actions), Docker, Kubernetes, Jira
- **Core:** Data Structures and Algorithms, Object-Oriented Programming, Operating Systems, Database Management Systems, System Design

ACHIEVEMENTS

- Made **10+ Personal Projects** in Web Development (Frontend + Backend) using Different Web Technologies.
- Solved **600+** coding problems across platforms [LeetCode](#) | [Codeforces](#)
- Finalist in 10+ hackathons, leading end-to-end development of AI, web, and cloud solutions, including **Smart India Hackathon (SIH)** and **IIC Hackathon**.

EDUCATION

B.Tech. (CSE) - 8.02 CGPA

Vellore Institute of Technology

📅 2023 – 2027

📍 Sehore, Bhopal

PROJECTS

SyncMetro

- **ReactJs | Material UI | PostgreSQL | Redis | Typescript**
- Built a full-stack metro booking platform with a modular **Microservices** architecture. Modeled the DMRC network as a weighted graph with a custom Dijkstra engine supporting 3 configurable routing strategies.
- Improved API response time by **97%** (4ms to 0.1ms) by integrating a **Redis** caching layer. Implemented cache invalidation strategies tied to real-time graph updates on a **Node.js** and **Express.js** backend.
- Built secure **JWT-based authentication** with OTP email verification, **bcrypt** password hashing, and **role-based access control (RBAC)**, ensuring secure access for both users and administrators. [Github](#)

Stock Broker

- **Next | GRPC | GraphQL | OpenSearch | AWS Lambda Search | PostgreSQL | Microservices**
- Replaced **REST** with **gRPC (HTTP/2, Protocol Buffers)**, reducing inter-service latency by **45%** (180 ms → 100 ms) and improving Risk Management Service throughput by **35%**
- Architected a high-throughput data ingestion pipeline using **OpenSearch** to index and query over **16MB** of raw CSV stock data, enabling sub-millisecond full-text searches and relieving downstream database overhead by offloading heavy lookup queries.
- Developed a **GraphQL Gateway** for the Order History Service, improving API efficiency through flexible, **single query** data fetching. [Github](#)

LeetSync

- **Javascript | Node.js | Manifest V3 | Notion SDK**
- Engineered a Chrome Extension using Manifest V3 that injects a custom UI into the LeetCode DOM. **Automated revision tracking**, eliminating manual data entry through seamless browser-level integration.
- Developed a Node.js backend service using **Notion SDK** and REST APIs. Captured and synced LeetCode problem **metadata to a structured Notion database**, reducing data entry time by **100%**. [Github](#)